

Polynomials – 4

Real roots: Descartes, Sturm and Newton's inequalities

Homework

8 сентября 2026 года

Theoretical minimum. Descartes' rule of signs; Sturm sequences and the exact count of real roots; Newton's inequalities and log-concavity; interlacing of the roots of P and P' ; the location of complex roots through the coefficients.

1. Prove Sturm's theorem and use it to determine the number of real roots of $x^5 - 3x - 1$ and of $x^4 + x^3 - x - 1$.
2. (VJIMC 1994) How many real roots does the polynomial $1 + x + \frac{x^2}{2} + \frac{x^3}{3} + \dots + \frac{x^n}{n}$ have?
3. (CIIM 2018) Let $\alpha < 0 < \beta$ and $f(x) = x(x - \alpha)(x - \beta)$. Let S be the set of real s for which $f(x) - s$ has three distinct real roots, and for $s \in S$ let $p(s)$ be the product of the smallest and the largest root. Determine the smallest possible value of $p(s)$ for $s \in S$.
4. (CIIM 2012) Let $n \geq 2$ and $p(x) = x^n + a_{n-1}x^{n-1} + \dots + a_0$ be real. Show that if $(x - 1)^{k+1}$ divides $p(x)$ for some positive integer k , then $\sum_{j=0}^{n-1} |a_j| > 1 + \frac{2k^2}{n}$.
5. (SEEMOUS) Let P be a real polynomial of degree five whose graph has three inflection points lying on one straight line. Compute the ratios of the areas of the bounded regions between this line and the graph.
6. (IMC) Suppose the graph of a polynomial of degree 6 is tangent to a straight line at three points A_1, A_2, A_3 , with A_2 between A_1 and A_3 .
 - a) Prove that if $|A_1A_2| = |A_2A_3|$, then the two areas bounded by the segments and the graph are equal.

- b) With $k = |A_2A_3|/|A_1A_2|$ and K the ratio of the corresponding areas, prove that $\frac{2}{7}k^5 < K < \frac{7}{2}k^5$.
7. (VJIMC 1998) Show that all complex roots of $P(z) = a_0z^n + a_1z^{n-1} + \cdots + a_{n-1}z + a_n$, where $0 < a_0 < a_1 < \cdots < a_n$, satisfy $|z| > 1$.
8. (MIPT) Let all roots, real and complex, of a real polynomial $p(x)$ have modulus greater than one. Prove that there is a real polynomial $q(x)$ divisible by $p(x)$ whose constant term exceeds in absolute value the sum of the absolute values of all its other coefficients.
9. Prove that between two consecutive roots of P' there is at most one root of a real-rooted P , and that for real-rooted P and every real c the polynomial $P + cP'$ is again real-rooted with roots interlacing those of P .
10. (Karasev 5.11) Let $a_1 < a_2 < \cdots < a_n$ and $b_1 < b_2 < \cdots < b_n$ be two sets of rational numbers. Prove that the equation $e^{a_1x} + \cdots + e^{a_nx} = e^{b_1x} + \cdots + e^{b_nx}$ has at most n real roots, and explain why this is Descartes' rule in disguise.